4.1 基本触发器的逻辑符号与输入波形如图P4.1所示。试作出Q、Q 的波形。

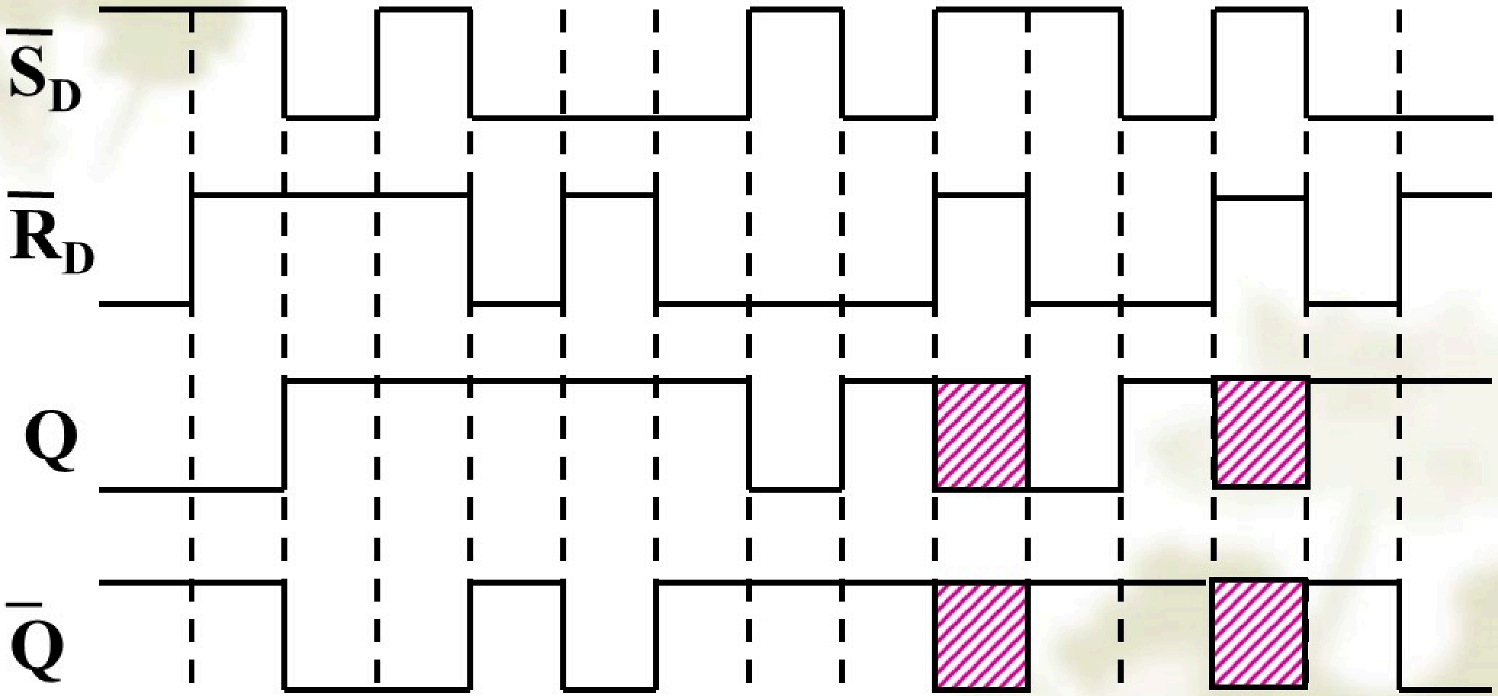

图 P5.1

4.3 对于图P4.3电路,试导出其特征方程并说明对A、B的取值有无约束条件。

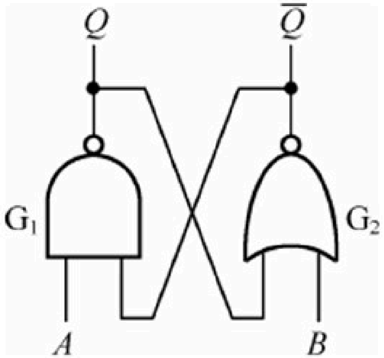

图 P4.3

解: (1)列功能表　　(2) 特征方程

A　B	Q^{n+1}
0　0	1
0　1	1
1　0	Q^n
1　1	1

A \ BQ^n	00	01	11	10
0	1	1	1	1
1	0	1	1	1

$$Q^{n+1} = \overline{A} + B + Q^n$$

(3) 对A、B的取值无约束条件。

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4.4试写出图 P4.4 触发器电路的特征方程。

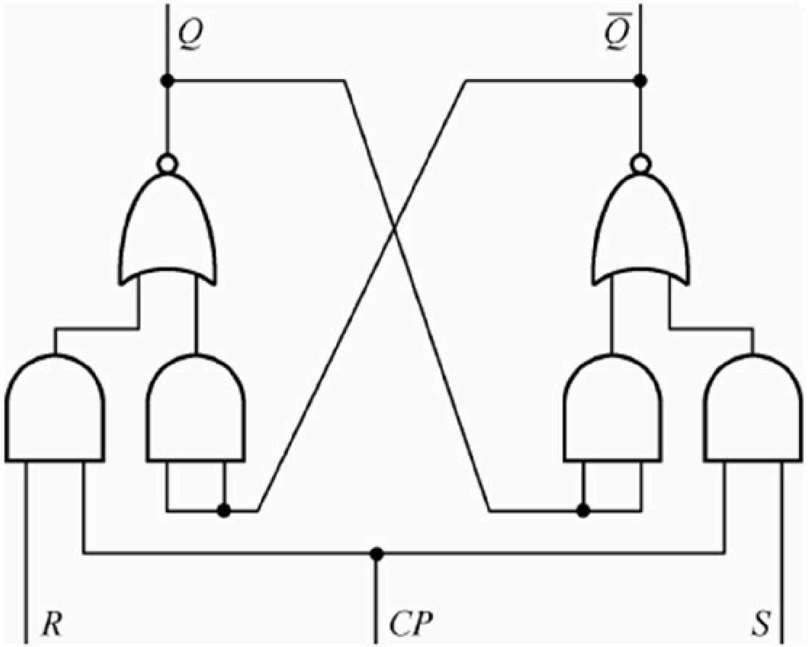

图 P4.4

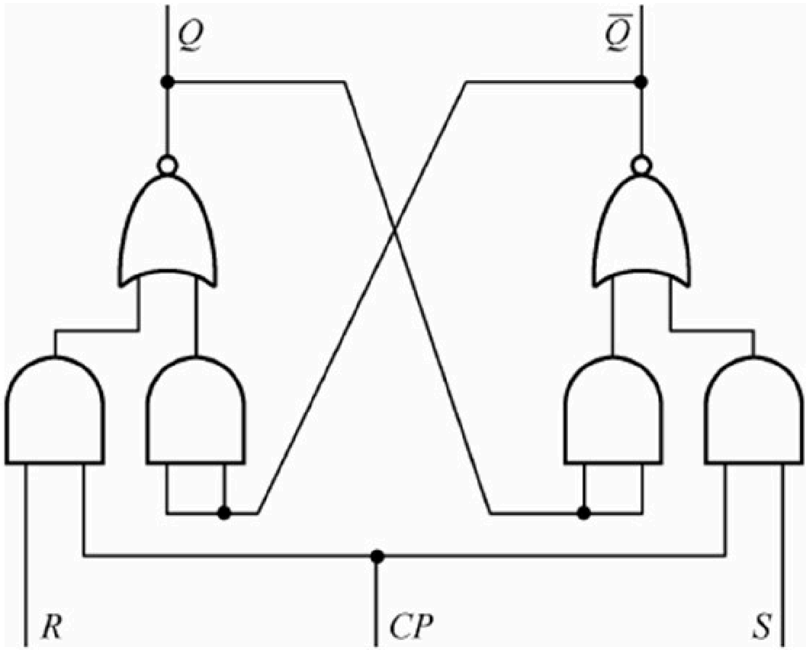

图 P4.4

CP=0时， $Q^{n+1}=Q^n$

CP=1时， $\begin{cases} Q^{n+1}=S+\bar{R}Q^n \\ SR=0 \end{cases}$

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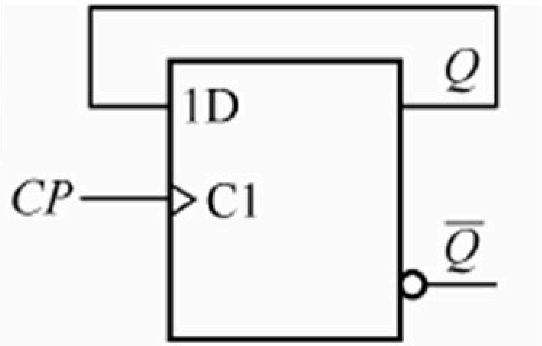
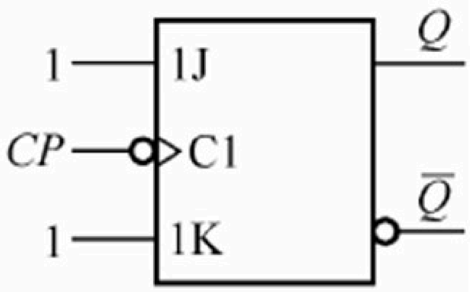
4.5 试写出图 P4.5 各触发器的次态方程。



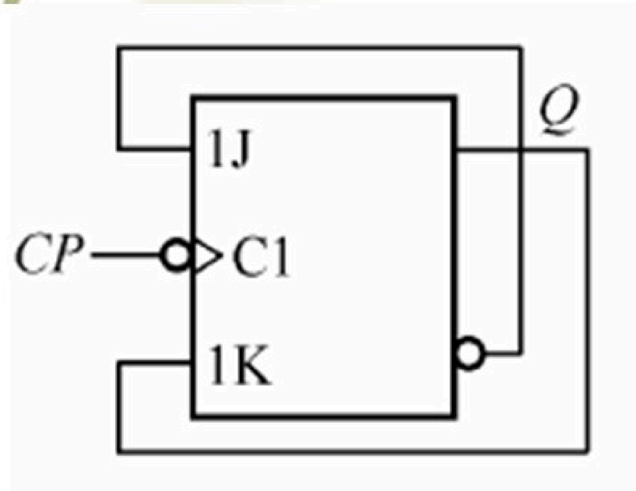
(b)



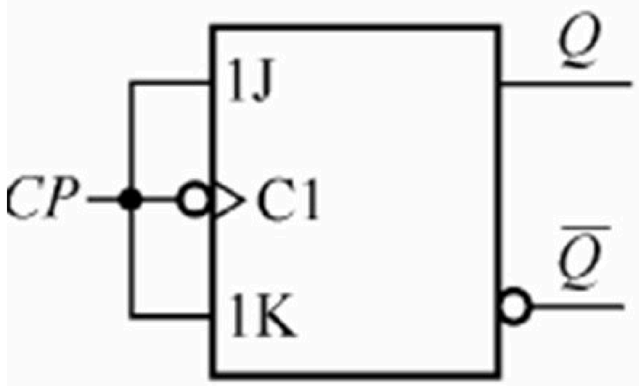
(d)

$$Q^{n+1}=D\cdot CP\uparrow$$
$$=[\overline{Q^n}]\cdot CP\uparrow$$

$$Q^{n+1}=\left[J\overline{Q^n}+\overline{K}Q^n\right]CP\downarrow$$
$$=\left[1\overline{Q^n}+\overline{1}Q^n\right]CP\downarrow$$
$$=\left[\overline{Q^n}\right]CP\downarrow$$



(e)

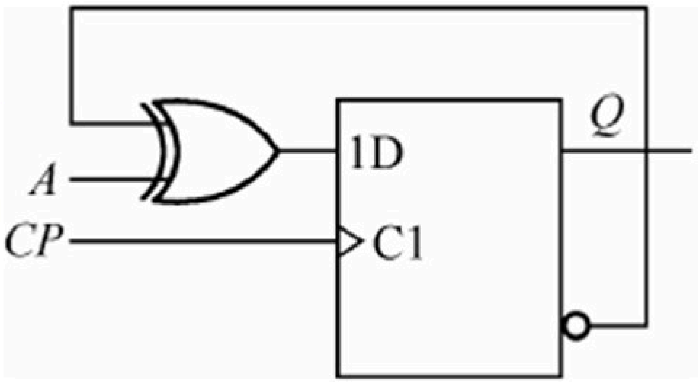


(g)

$$Q^{n+1}=\left[J\overline{Q^n}+\overline{K}Q^n\right]CP\downarrow$$
$$=\left[\overline{Q^n}\,\overline{Q^n}+\overline{Q^n}Q^n\right]CP\downarrow$$
$$=\left[\overline{Q^n}\right]CP\downarrow$$

$$Q^{n+1}=\left[J\overline{Q^n}+\overline{K}Q^n\right]CP\downarrow$$
$$=\left[CP\overline{Q^n}+\overline{CP}Q^n\right]CP\downarrow$$
$$=\left[1\overline{Q^n}+\bar{1}Q^n\right]CP\downarrow$$
$$=\left[\overline{Q^n}\right]CP\downarrow$$

4.8画出图P4.8中Q端的波形。设初态为“0”。



解：特征方程为：

$$Q^{n+1}=D\cdot CP\uparrow$$
$$=[\overline{Q}^{n}\oplus A]\cdot CP\uparrow$$

A=0时，翻转；
A=1时，保持。

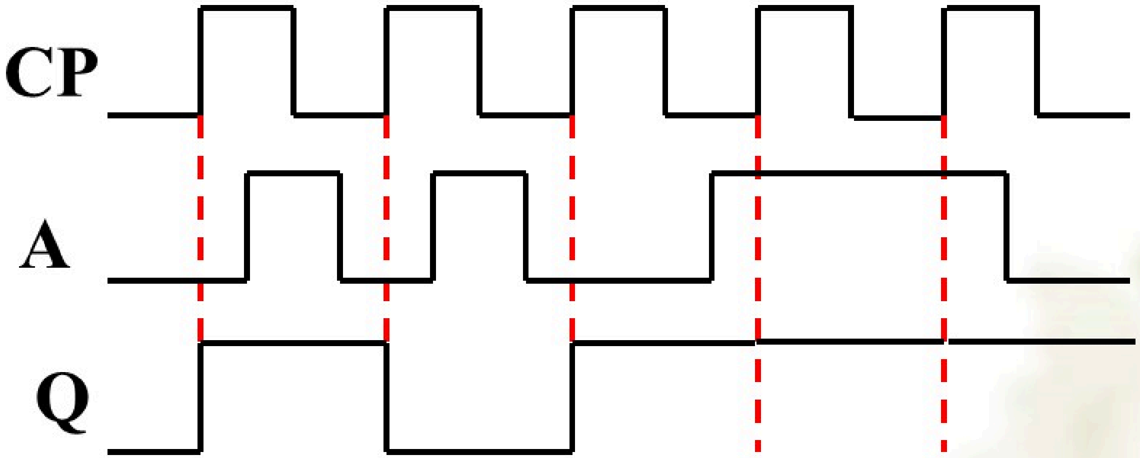


图 4.8

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4.11画出图P4.11电路中 Q_1和 Q_2 的波形。

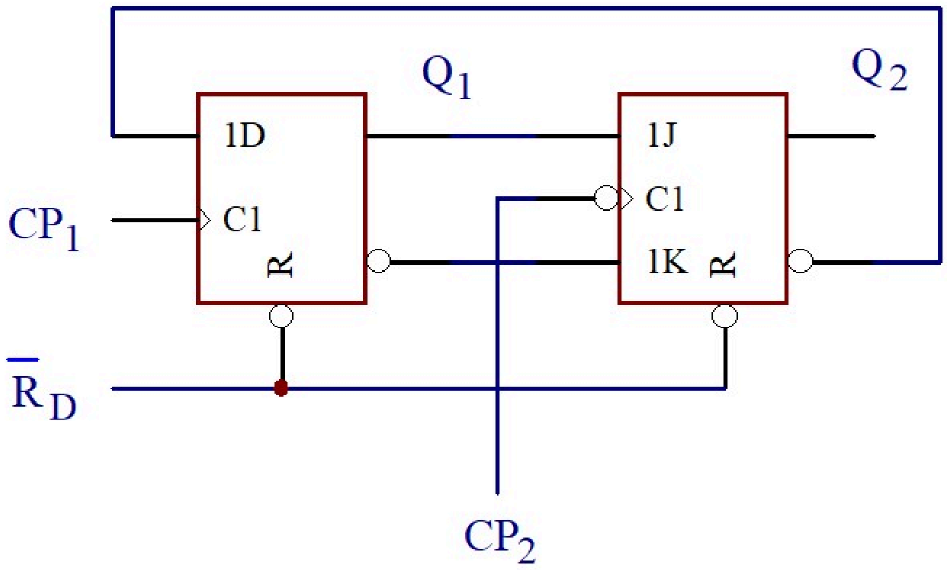

图 4.11

解：特征方程为：

$$Q_1^{n+1}=[\ \overline{Q}_2^{\,n}]CP_1\uparrow$$

$$Q_2^{n+1}=[Q_1^{\,n}]CP_2\downarrow$$

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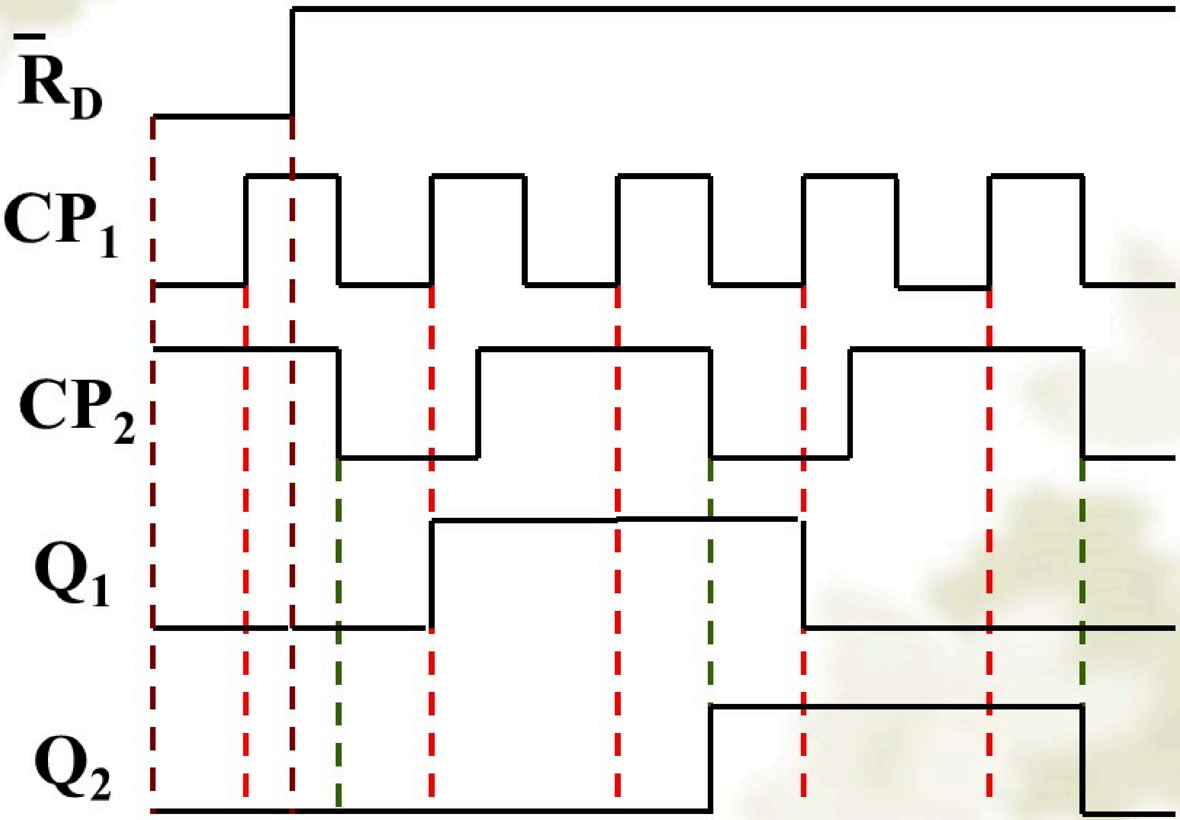
$Q_1^{n+1}=[\ \overline{Q}_2^{\,n}]CP_1\uparrow \qquad Q_2^{n+1}=[Q_1^{\,n}]CP_2\downarrow$



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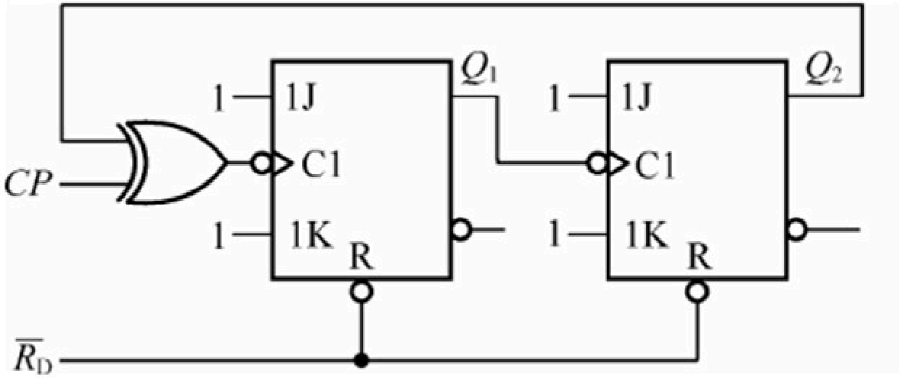
4.17 试作出图P4.17电路中Q_1、Q_2的波形。



$$Q_1^{n+1}=[\ \overline{Q_1^n}\ ]\cdot(CP\oplus Q_2)\downarrow$$

Q_2=0时

$$Q_1^{n+1}=\left[\overline{Q_1^n}\right]\cdot(CP\oplus 0)\downarrow$$

$$=\left[\overline{Q_1^n}\right]\cdot(CP)\downarrow$$

Q_2=1时

$$Q_1^{n+1}=\left[\overline{Q_1^n}\right]\cdot(CP\oplus 1)\downarrow$$

$$=\left[\overline{Q_1^n}\right]\cdot(\overline{CP})\downarrow$$

$$=\left[\overline{Q_1^n}\right]\cdot(CP)\uparrow$$

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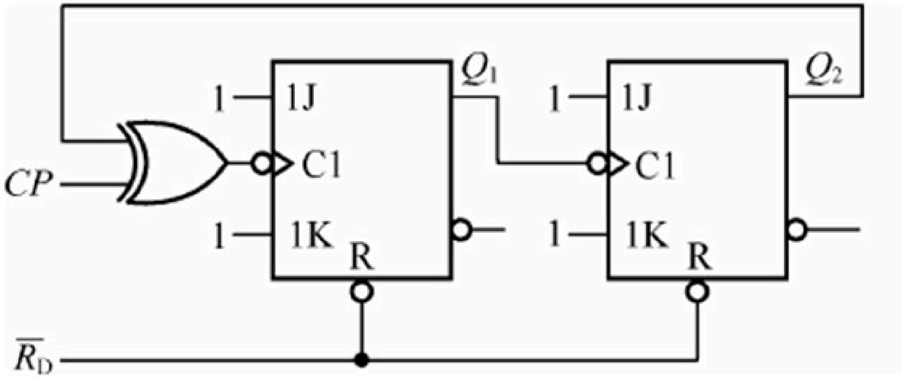
4.17 试作出图P4.17电路中Q_1、Q_2的波形。

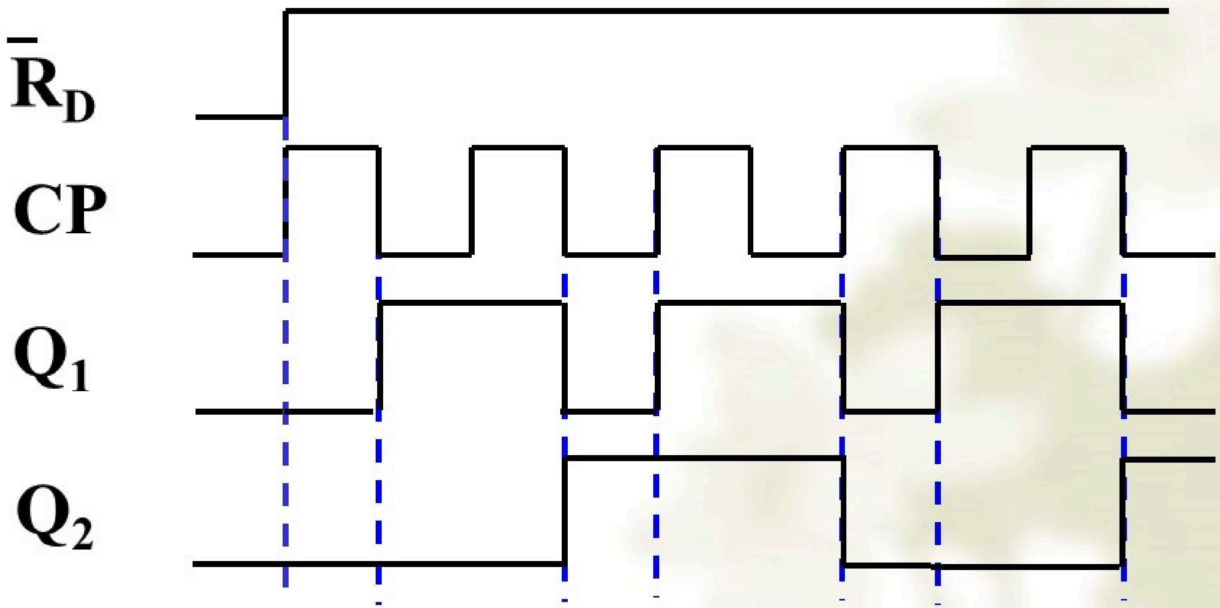


$$Q_1^{n+1}=[\ \overline{Q_1^n}\ ]\cdot(CP\oplus Q_2)\downarrow$$

Q_2=0时，CP↓

Q_2=1时，CP↑

$$Q_2^{n+1}=[\ \overline{Q_2^n}\ ]\cdot Q_1\downarrow$$



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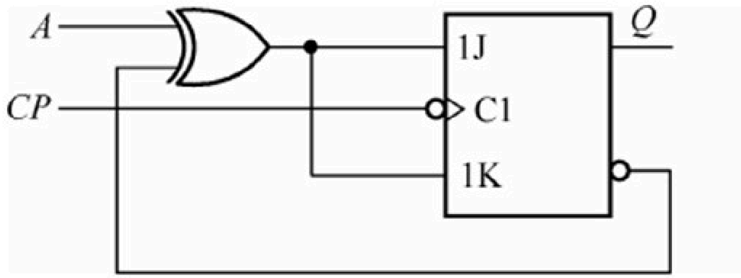
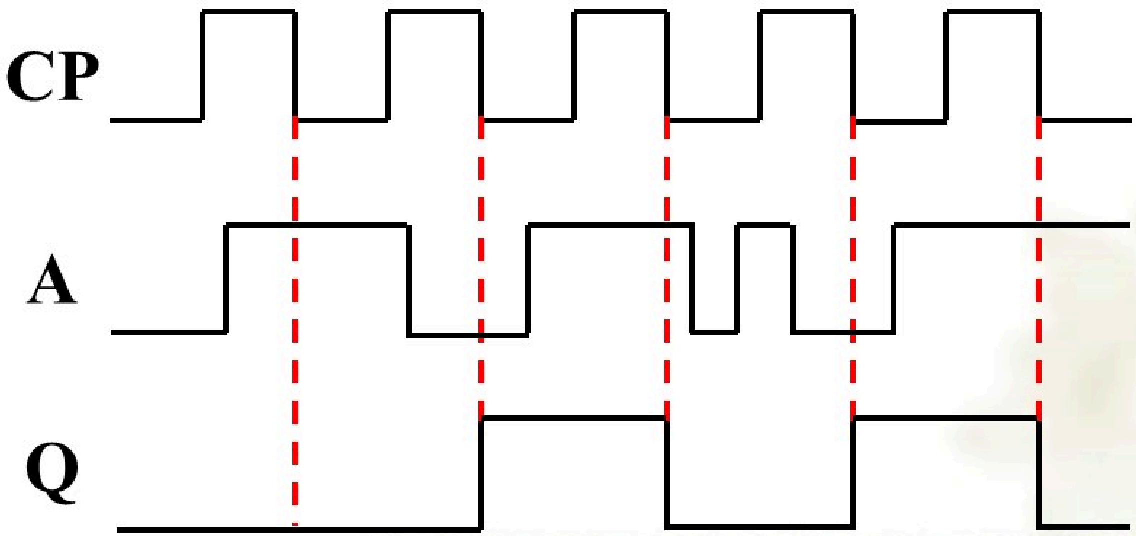
4.19已知电路如图P4.19，试作出Q端的波形。设Q的初态为“0”。



$$Q^{n+1}=\left[J\overline{Q^n}+\overline{K}Q^n\right]CP\downarrow$$

$$=\left[\left(A\oplus\overline{Q^n}\right)\overline{Q^n}+\left(\overline{A\oplus\overline{Q^n}}\right)Q^n\right]CP\downarrow$$

$$=\left[\overline{A}\right]CP\downarrow$$



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